

Tighter Control of Quantum Cascade Laser Frequency Combs with a Phase-Locked Loop

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Quantum Cascade Laser Frequency Combs

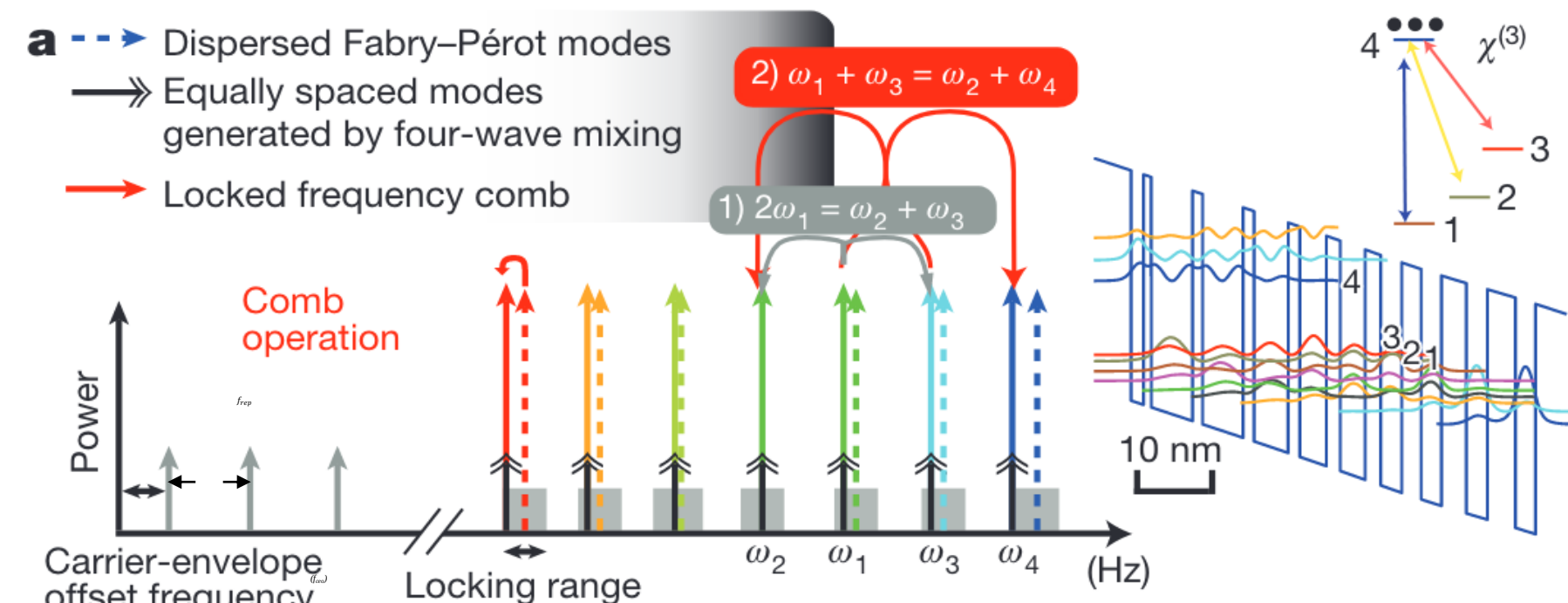


Fig. 1: QCL-FC operation¹

Advantages:

- Mid-IR
- Self-starting comb generation

Disadvantages:

- Hard to reproduce
- Unstable f_{rep} and f_{ceo}

Coherent Control with RF injection

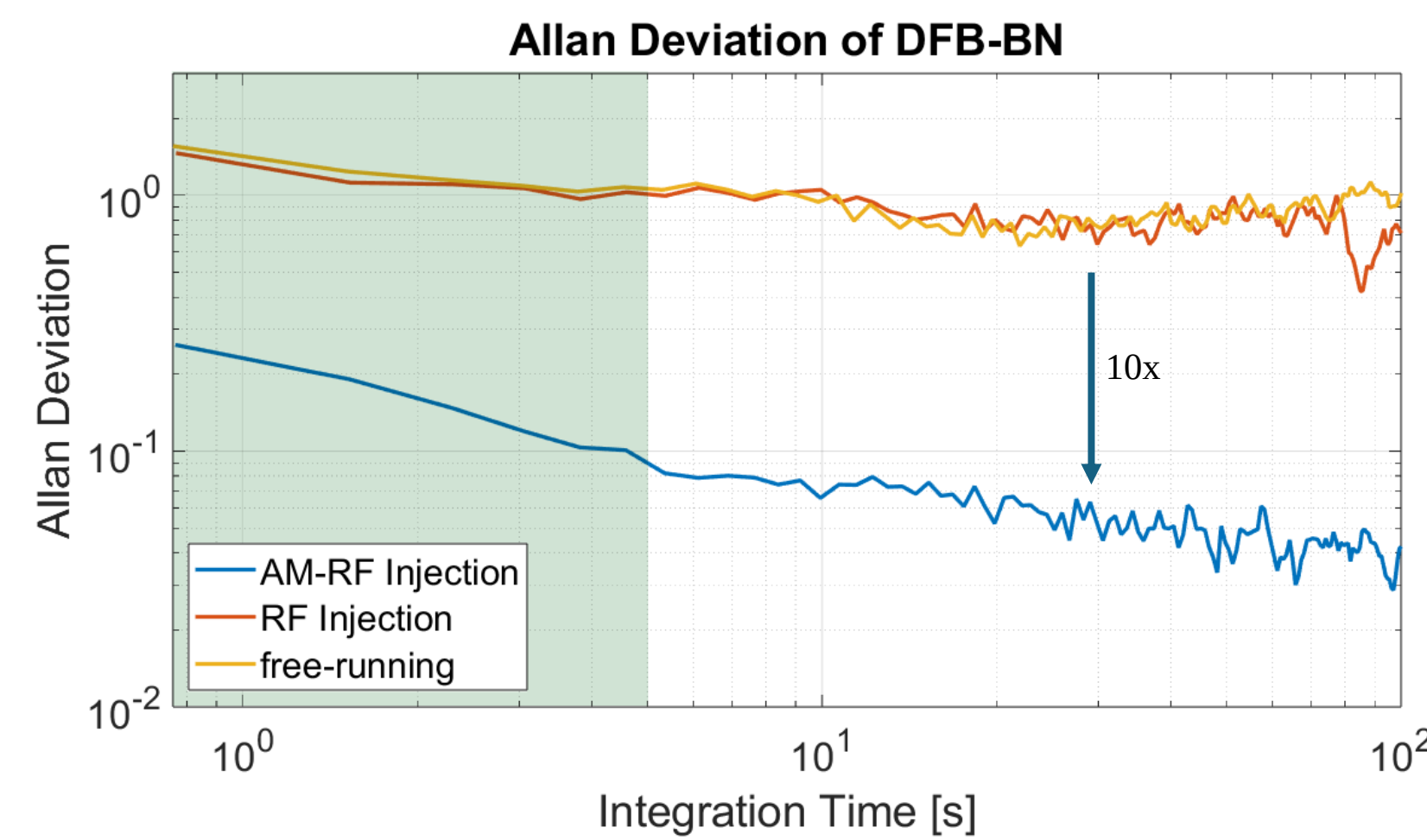
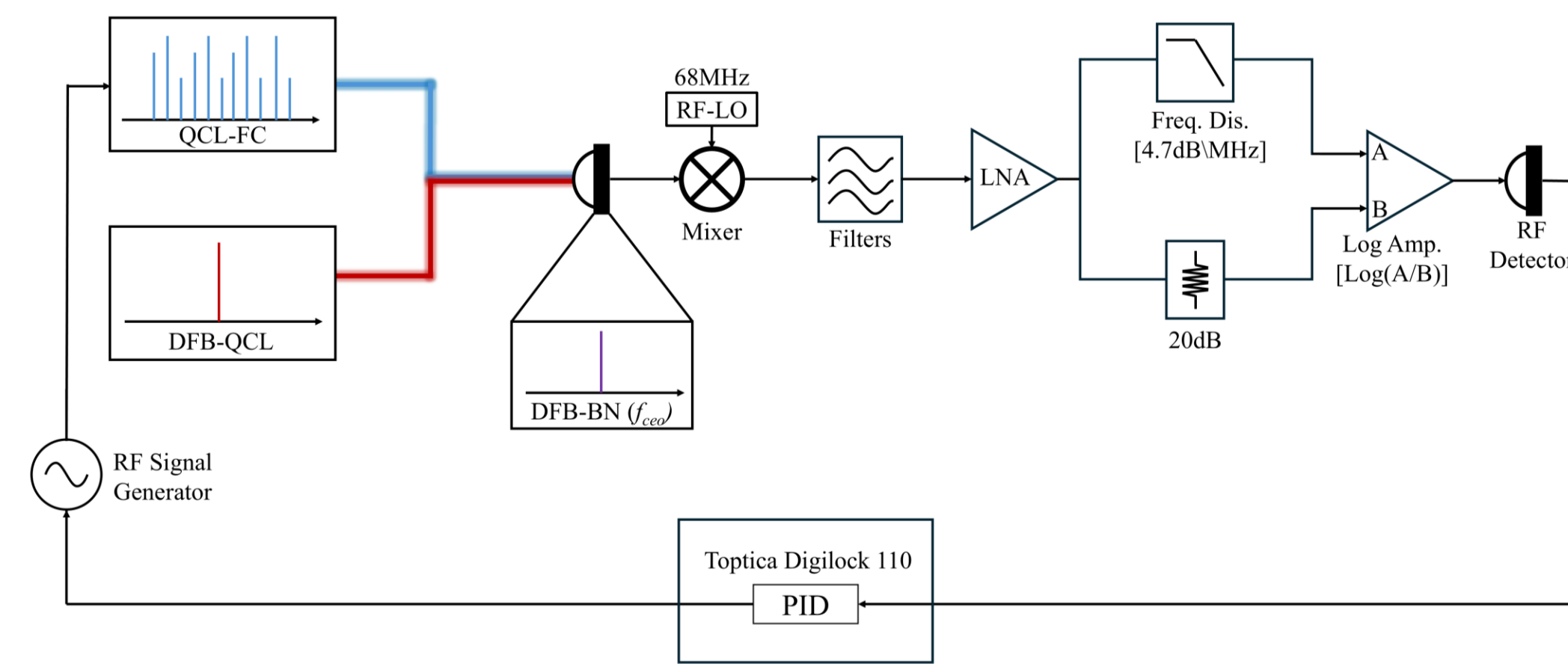


Fig. 2: Coherent control of f_{ceo} and f_{rep} with amplitude-modulated RF injection. Fig. 2a (top) is the locking schematic. Fig. 2b (bottom) demonstrates successful locking of f_{ceo}

By adding amplitude modulation to the RF injection signal, Allan Deviation decreases by an order of magnitude.

Now, we are targeting improvement of the green portion of the graph (or even beyond to integration times of ms) requiring a faster feedback system.

Phase-Locked Loop Implementation Feasibility

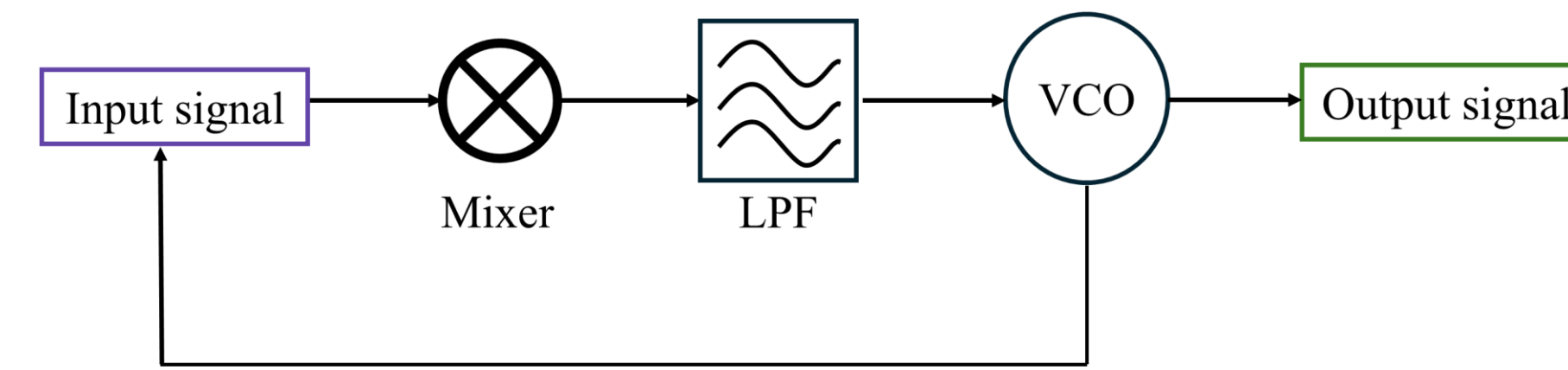


Fig. 3: A typical PLL circuit, with a phase detector, a low-pass filter, and a voltage-controlled oscillator

A phase-locked loop (PLL) is a control loop outputting a signal whose phase and frequency match that of a reference input signal. Its functionality is the same as the frequency discriminator circuit we used previously, but its output signal can match faster fluctuations in the input signal.^{2,3,4}

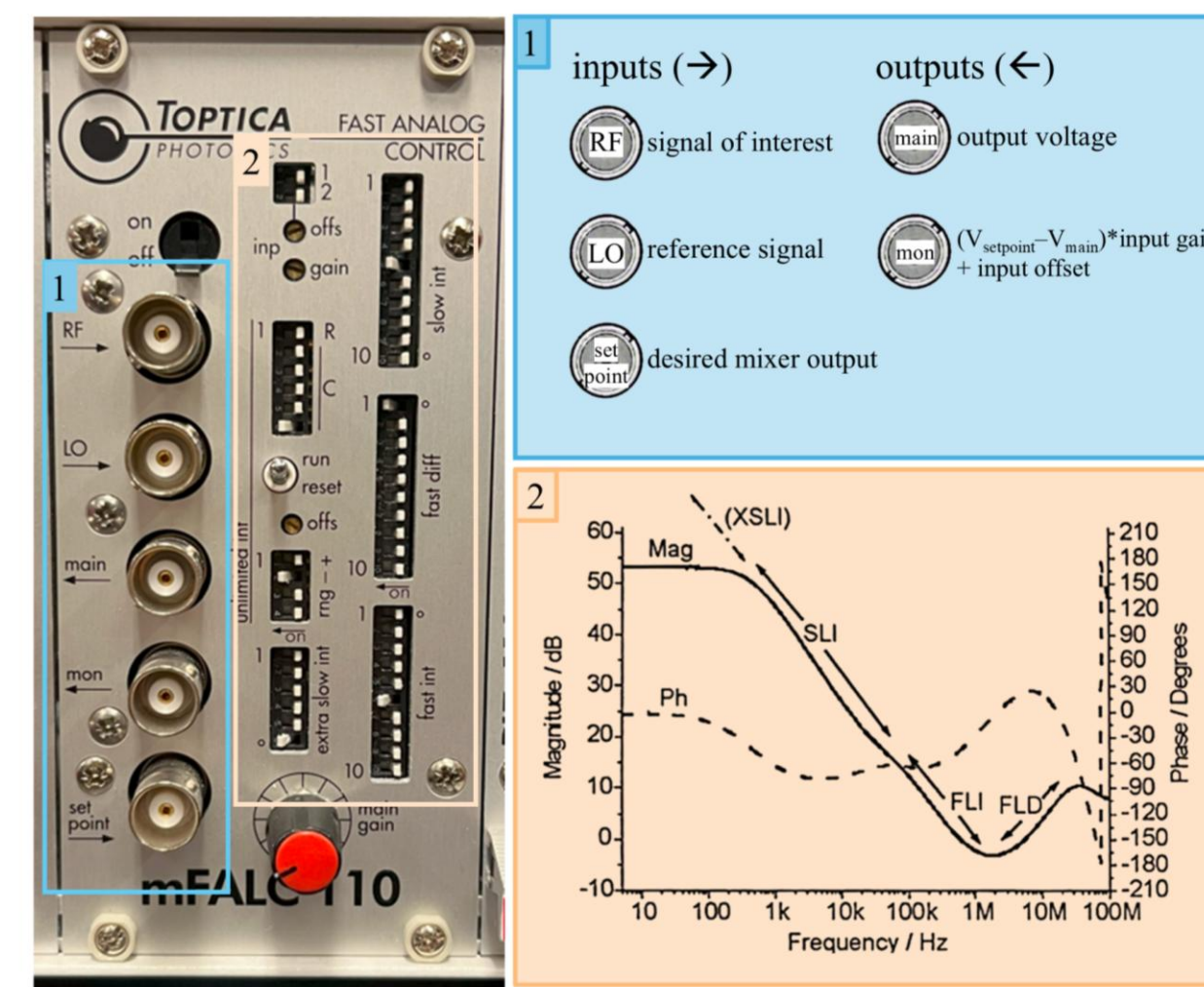


Fig. 4: The front panel of the Toptica mFALC110, our PLL circuit unit

Understanding our PLL circuit unit:

Box 1: The 3 input and 2 output ports

Box 2: The pins controlling the module's PID controller. The Bode plot in orange shows which pin sets affect which input signal frequencies

Determining the holding range of the PLL module in order to inform our choice of LO frequency and control loop design:

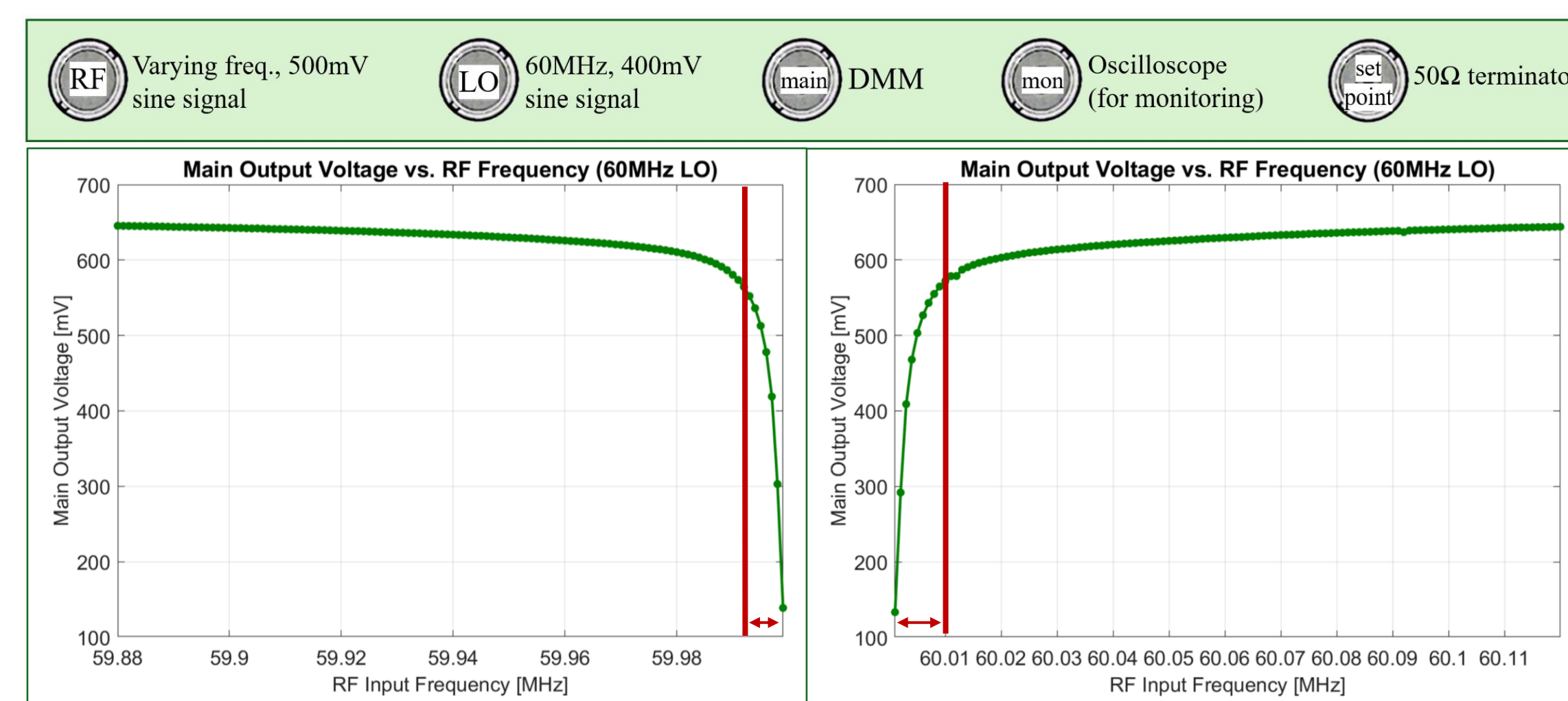


Fig. 5: Frequency response of the PLL main output. The holding range is indicated in red.

The PLL module's holding range (~10kHz, Fig. 5) is smaller than the typical drift of the DFB-BN, necessitating a slower control loop to accompany the PLL circuit loop.

Proposed Implementation

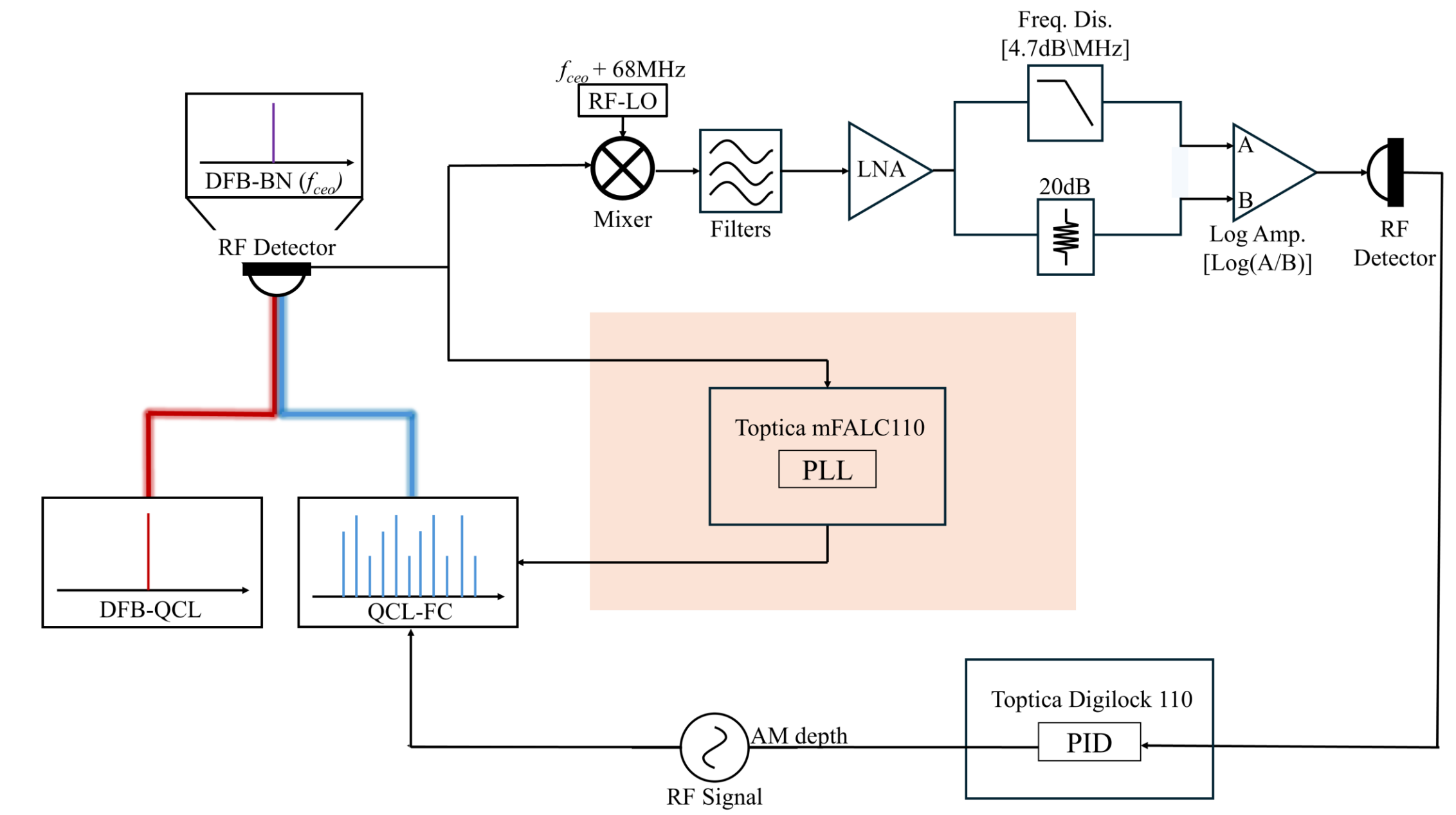


Fig. 6: Proposed schematic for locking the QCL-FC with two control loops

1. Slow locking loop: The loop with the frequency discriminator circuit controlling amplitude modulation depth of the injected RF signal.
2. Fast locking loop: Provides a tighter lock of f_{ceo} using a phase-locked loop. The output of the PLL modulates the laser driver current.

The laser driver has a maximum modulation bandwidth of 2-3MHz, while the amplitude modulation bandwidth on the RF signal generator goes up to 1MHz. Hence the choice of the fast loop to control the laser driver.

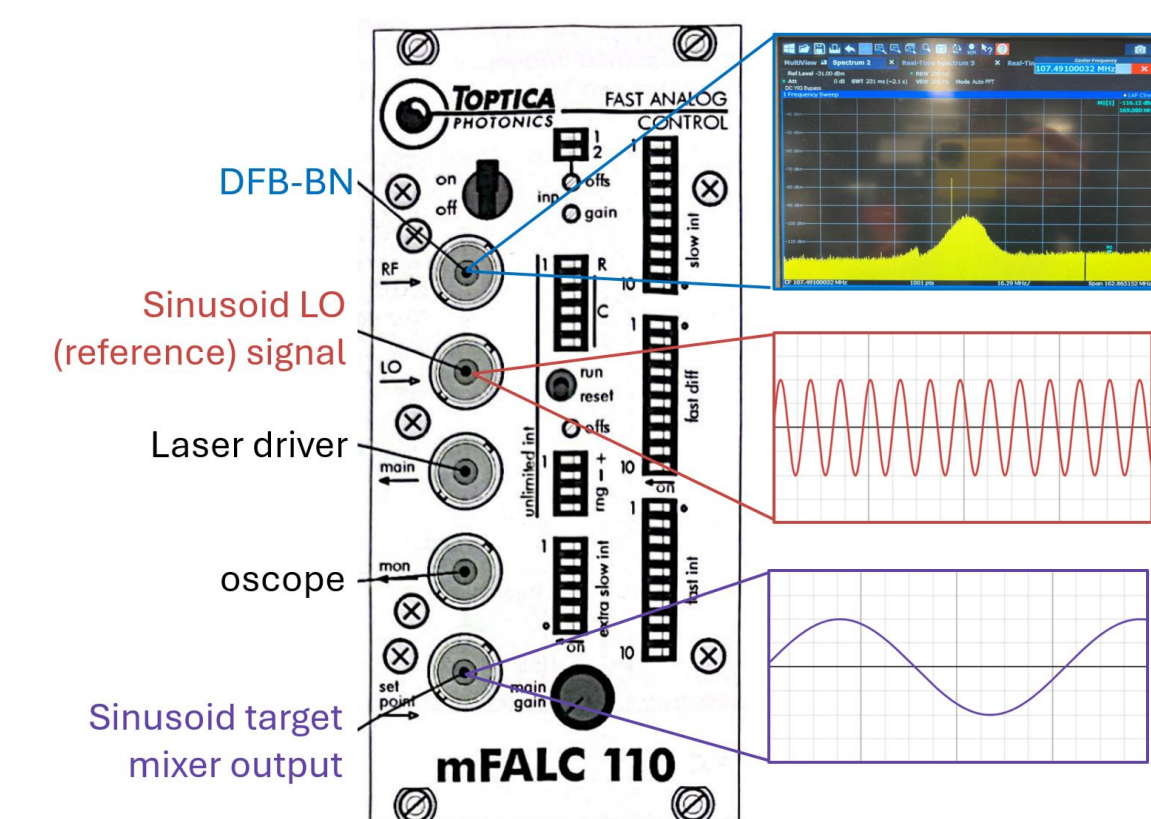


Fig. 7: Proposed setup of the PLL module

The module will be set up as such:

1. DFB-BN signal → RF input
2. LO signal → LO input
3. Laser driver ← Main output
4. Oscilloscope ← Mon output
5. Desired mixer output → Setpoint input

Conclusions, Future Directions

We have designed a double locking loop with the addition of a phase-locked loop to provide a tighter lock of f_{ceo} of a QCL-FC

Next steps: implementation, from which we hope will lead to lower Allan Deviation of the DFB-BN frequency for lower integration times

Acknowledgements

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References

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